Software Requirement Specifications

For RideNED

Version # 01

Prepared by Group ID: A4

Raazia Imran (SE-24009)
Muzammil Ali (SE-24034)
Rameez Kamran (SE-24047)
Abdullah Zia (SE-24048)

14 April 2025

Department of Software Engineering
NED University of Engineering and Technology

Revision History

Req ID	Date	Reason for Change	Version

Contents

1 Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the **RideNED** application. This document lists the functional and non-functional requirements and creates a clear understanding between the stakeholders and the developers regarding the system features.

1.2 Document Convention

This document follows the IEEE standard for Software Requirement Specifications. Requirements are prioritized as High, Medium, or Low.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Developers:** To understand the features to be implemented.
- **Testers:** To validate the system against requirements.
- **Project Managers:** To track progress and scope.
- **Stakeholders:** To verify that their needs are met.

1.4 Product Scope

RideNED is a mobile application designed to facilitate the transportation system at NED University. It digitizes the process of tracking points, paying fees, and managing passes.

2 Overall Description

2.1 Product Perspective

The product is not independent and not a standalone system as it is a part of the larger NED University's transport management system. It relies on:

- The university's existing student authentication system.
- Google Maps services for location tracking.
- Hardware scanners for marking point attendance digitally.

2.2 Product Functions

Note: Detailed functions are listed in Section 3 (System Features).

2.3 User Classes and Characteristics

The following are the main users of this application:

2.3.1 NED University Students

The NED student is an entity that uses the mobile application to access transportation-related services.

- **Capabilities:** Track live point locations, report complaints, report lost/found items, book seats, access digital bus passes, and pay point fees.
- **Prerequisites:** Must have university account credentials. Expected to be familiar with basic mobile navigation, online payments, and QR codes.
- **Permissions:** Must allow microphone access (voice assistant) and location services.

2.3.2 NED Admin

The admin has access to the complete features of the application.

- **Responsibilities:** Manage overall operations, update routes, monitor locations, manage lost/found reports, and upload trip data.
- **Skills:** Efficient in managing digital dashboards, data organization, and responding to user requests accurately.

2.4 Operating Environment

The project is semi-automated. It automates features like live tracking and digital passes but requires administrative tasks (managing routes, handling complaints) ensuring human oversight for critical decisions.

2.5 User Documentation

Student Interface:

- Install app and grant permissions (location, storage).
- Register using university credentials and verify via OTP (Gmail).
- Access dashboard for tracking, fees, and forms.

Admin Interface:

- Register as admin using valid university credentials and verify via OTP.
- Access admin panel to manage routes, complaints, statistics, and transactions.

2.6 Assumptions and Dependencies

- Students possess valid university-issued credentials.
- Points are operational based on provided schedules.
- Points are equipped with "Eyoyo USB QR Code Scanners."
- Google Maps services are available for tracking.
- Students use Android/iOS smartphones with GPS enabled.

3 System Features

3.1 University Point Tracking System

Author: Rameez Kamran [SE-24047]

3.1.1 Description and Priority

This feature helps students view the real-time location of university points on a map using GPS. It addresses the issue of uncertainty regarding arrival times.
Priority: High

3.1.2 Response Sequence

- User logs in and selects “Routes” from the dashboard.
- System displays list of points (R1, R2, etc.).
- User selects a route; system fetches GPS location from the tracker.
- System displays live location on the map, refreshing continuously.

3.1.3 Functional Requirements

FT REQ-1 Users can sort points by route number, bookmarked routes, or ‘optimized suggestion‘ based on device GPS.

FT REQ-2 User can view the list of stops for a point with exact locations on the map.

FT REQ-3 User can bookmark up to three points for dashboard access. Oldest bookmark is replaced if limit exceeded.

FT REQ-4 ‘Notify Me‘ option sends a message when a point is 5 minutes away from a selected stop.

FT REQ-5 Home screen widget displays live location.

FT REQ-6 Map view defaults to roadmap but supports satellite view.

FT REQ-7 User can share point location/ETA via WhatsApp (includes map link).

FT REQ-8 ETA for stops updates automatically based on traffic data.

FT REQ-9 User can view driver details (name, picture, route).

FT REQ-10 Voice input support for visually impaired users (e.g., “Where is route 3?”).

FT REQ-11 In-built voice assistant requires microphone permission.

3.1.4 Non-Functional Requirements

FT REQ-4.1 Popups disappear automatically in 10 seconds.

FT REQ-5.1 Only one tracking widget allowed at a time.

FT REQ-5.2 Widget does not auto-update to save battery (manual refresh required).

FT REQ-7.1 Driver image must be passport-sized, JPEG/PNG, blue background, $<$ 1MB.

FT REQ-9.1 Driver info fetch time must not exceed 10 seconds.

3.2 Report Lost/Found Items

Author: Rameez Kamran [SE-24047]

3.2.1 Description and Priority

A centralized platform for reporting and recovering lost items on university points.
Priority: Medium

3.2.2 Response Sequence

- User selects "Report Lost/Found Item".
- User fills form (description, route, photo).
- Item is posted to Lost and Found Board.
- Matched users connect via in-app chat. Item removed when marked "Returned".

3.2.3 Functional Requirements

FT REQ-12 Form requires item name, description, date, route, and optional image.

FT REQ-13 User marks item as "claimed" by describing a hidden detail.

FT REQ-14 Found items are public; Lost items are private until a match is found.

3.2.4 Non-Functional Requirements

FT REQ-14.1 Contact numbers hidden; chat history refreshes 48 hours after claim.

FT REQ-12.1 Description limit 150 chars; Image limit 1MB.

FT REQ-12.5 Anti-spam: Max 1 report per day.

3.3 Digital Point Fee Payment

Author: Raazia Imran [SE-24009]

3.3.1 Description and Priority

Allows students to pay point fees via E-wallet or Bank Transfer and upload proof for admin verification.
Priority: High

3.3.2 Response Sequence

- Student selects payment method.
- System checks eligibility.
- Student pays and uploads screenshot/receipt.
- Admin verifies; if approved, student proceeds to pass generation.

3.3.3 Functional Requirements

FT REQ-1 Supports E-wallet (EasyPaisa/JazzCash) and Bank Transfer (NBP/HMB).

FT REQ-2 Wallet: Requires Transaction ID and screenshot.

FT REQ-3 Bank: Displays account details; requires Ref Number and receipt.

FT REQ-4 Admin interface for approving/rejecting payments.

FT REQ-5 "Proceed to Pass Generation" enabled upon approval.

FT REQ-6 Failed payments logged with error codes.

3.3.4 Non-Functional Requirements

FT REQ-1.1 Only registered on-campus students are eligible.

FT REQ-2.1 Proofs stored in secure cloud bucket (AES-256 encrypted).

FT REQ-4.1 Verification completion target: 2 hours (working hours).

3.4 Digital Point Pass

Author: Raazia Imran [SE-24009]

3.4.1 Description and Priority

Generates a QR-based monthly pass tied to Student ID, replacing manual checks.
Priority: High

3.4.2 Functional Requirements

FT REQ-7 access allowed only if semester fee is “Paid”.

FT REQ-8 Mandatory fields: ID, Name, Phone (OTP), Dept, Batch, Photo.

FT REQ-10 Pass contains details and encrypted QR code.

FT REQ-12 OS-level restrictions on screenshots/recording.

FT REQ-14 Countdown timer for expiry; renewal allowed with < 7 days remaining.

FT REQ-15 History of past passes and payments.

3.4.3 Non-Functional Requirements

FT REQ-1.2 Daily jobs check validity and send expiry reminders (3 days prior).

FT REQ-10.1 Unique QR hash regenerated every 24 hours.

FT REQ-11.1 Pass accessible offline.

3.5 Point Pre-Booking System

Author: Muzammil Ali [SE-24034]

3.5.1 Description and Priority

Allows booking of Department-Specific or General trips with seat management.
Priority: Medium

3.5.2 Functional Requirements

FT REQ-1 Trips categorized as “Department-Specific” or “General”.

FT REQ-3 System filters trips based on student department.

FT REQ-6 Support for trips requiring special approval (e.g., HOD).

FT REQ-9 Prevention of overlapping bookings.

FT REQ-10 Live seat counters.

3.5.3 Non-Functional Requirements

FT REQ-1.1 100% accurate filtering of eligibility.

FT REQ-4.1 Consistent color-coded badges for trip categories.

3.6 Driver Complaint System

Author: Muzammil Ali [SE-24034]

3.6.1 Description and Priority

Streamlines reporting of unsafe driving or behavior with anonymous options.
Priority: Medium

3.6.2 Functional Requirements

FT REQ-14 Auto-fills trip details (Vehicle, Driver).

FT REQ-16 Anonymous submission option (identity hidden from driver).

FT REQ-17 Progress timeline (Under Review, Resolved).

FT REQ-20 Auto-save drafts every 15 seconds.

FT REQ-21 Tone analyzer warns against abusive language.

3.6.3 Non-Functional Requirements

FT REQ-18.1 Media upload: JPG, PNG, MP3 (< 10MB).

3.7 Admin Dashboard

Author: Abdullah Zia [SE-24048]

3.7.1 Description and Priority

Control panel for managing points, passes, announcements, and statistics.
Priority: High

3.7.2 Functional Requirements

FT REQ-1 View and sort user complaints.

FT REQ-2 Generate user point usage reports (PDF).

FT REQ-6 Add/Remove routes and stops.

FT REQ-7 Cancel suspicious digital passes.

FT REQ-10 Emergency alert for route deviation.

FT REQ-17 Notification templates (Pass Expiring, Maintenance).

FT REQ-21 Password re-entry required for critical deletions.

3.7.3 Non-Functional Requirements

FT REQ-2.1 PDF generation < 5 seconds.

FT REQ-13.1 Admin changes reflect in < 5 seconds.

4 Other Non-Functional Requirements

4.1 Performance Requirements

The app must be optimally responsive across devices (smartphones, tablets). This includes fast loading times, smooth animations, and seamless transitions.

4.2 Database Requirements

A NoSQL database (e.g., MongoDB) will be used to handle real-time data, including:

- Booking data (Student IDs, Trip IDs).
- Trip data (Routes, Seat availability).
- Real-time bus location (GPS coordinates) and QR code data.

4.3 Safety Requirements

- Scanner cables must be insulated to prevent electrical issues.
- Secure connection between scanner and device to prevent data loss.
- Scanners must contain overload protection and auto-shutdown features for overheating.